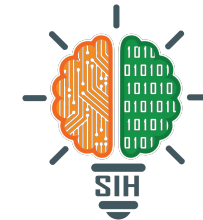




SMART INDIA HACKATHON 2026



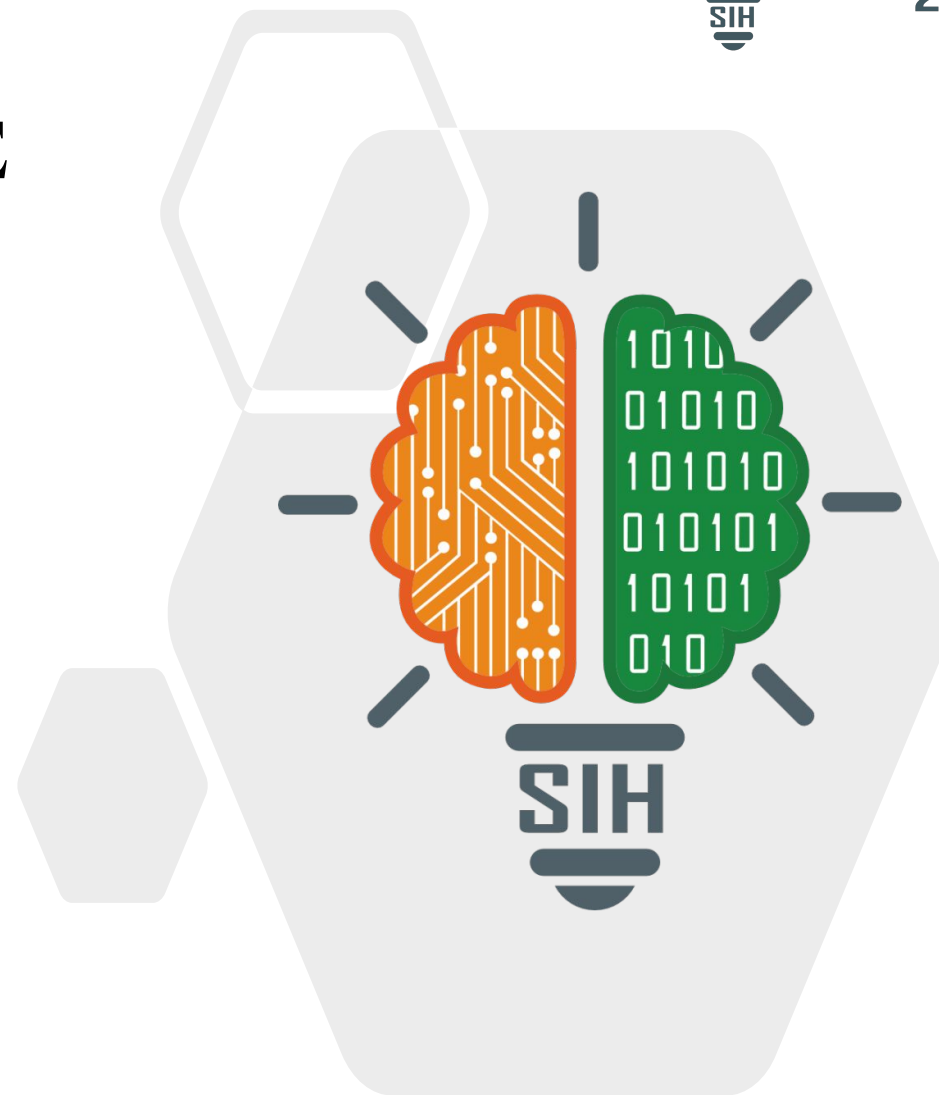
SMART INDIA
HACKATHON
2026

BLUE TRACE

- Problem Statement ID –SIH26143
- Problem Statement Title- SATELLITE + AIS

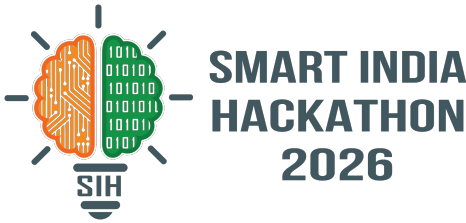
OIL SPILL

- Theme-DISASTER MANAGEMENT
- PS Category- Software
- Team ID-SIH26-A0H-T240





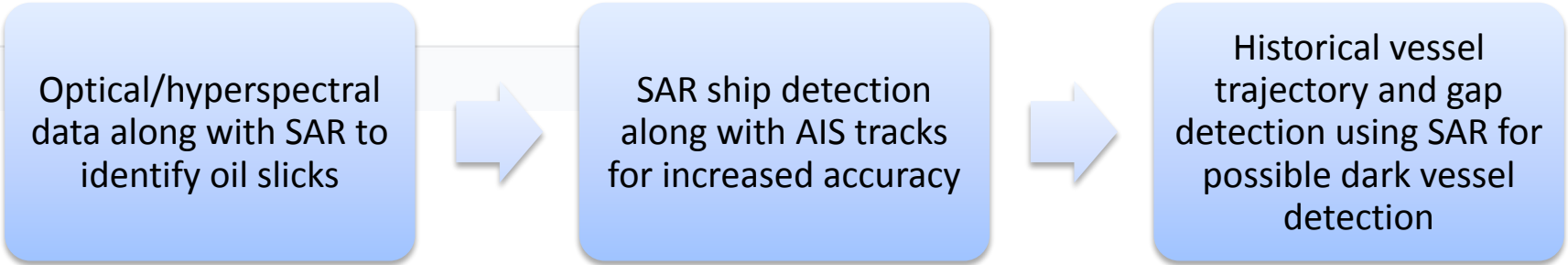
BLUE TRACE - SATELLITE + AIS OIL SPILL



STAGE	EXISTING LIMITATION	CONSEQUENCE
Detection	SAR dark spots ≠ always oil <i>(Look-alikes: low wind, algae, rain)</i>	False positives: Wasted operational resources, unnecessary and emergency dispatch, user alert fatigue.
Attribution	Spill-source trajectory prediction <i>has high uncertainty due to drift variables.</i>	Wrong source attribution: Misdirected liability, incorrect fines, and legal challenges against non-guilty operators.
Accountability	Heavy structural dependence on AIS tracking systems.	Dark vessels escape: Intentional disabling or spoofing of AIS allows non-compliant ships to evade detection.

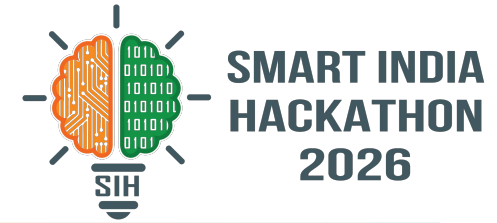
AIS BEHAVIOUR CHECKS

Stopping / speed anomaly	Unexpected stop or speed shift near the estimated source window.
Course / route anomaly	Sharp turn or route deviation inconsistent with the voyage pattern.
Loitering	Abnormally long slow movement inside or near the suspected source zone.
AIS gap	Missing transmissions near the critical time window; treated only as a clue.
Repeated behaviour	Similar anomalies appearing across historical vessel tracks.

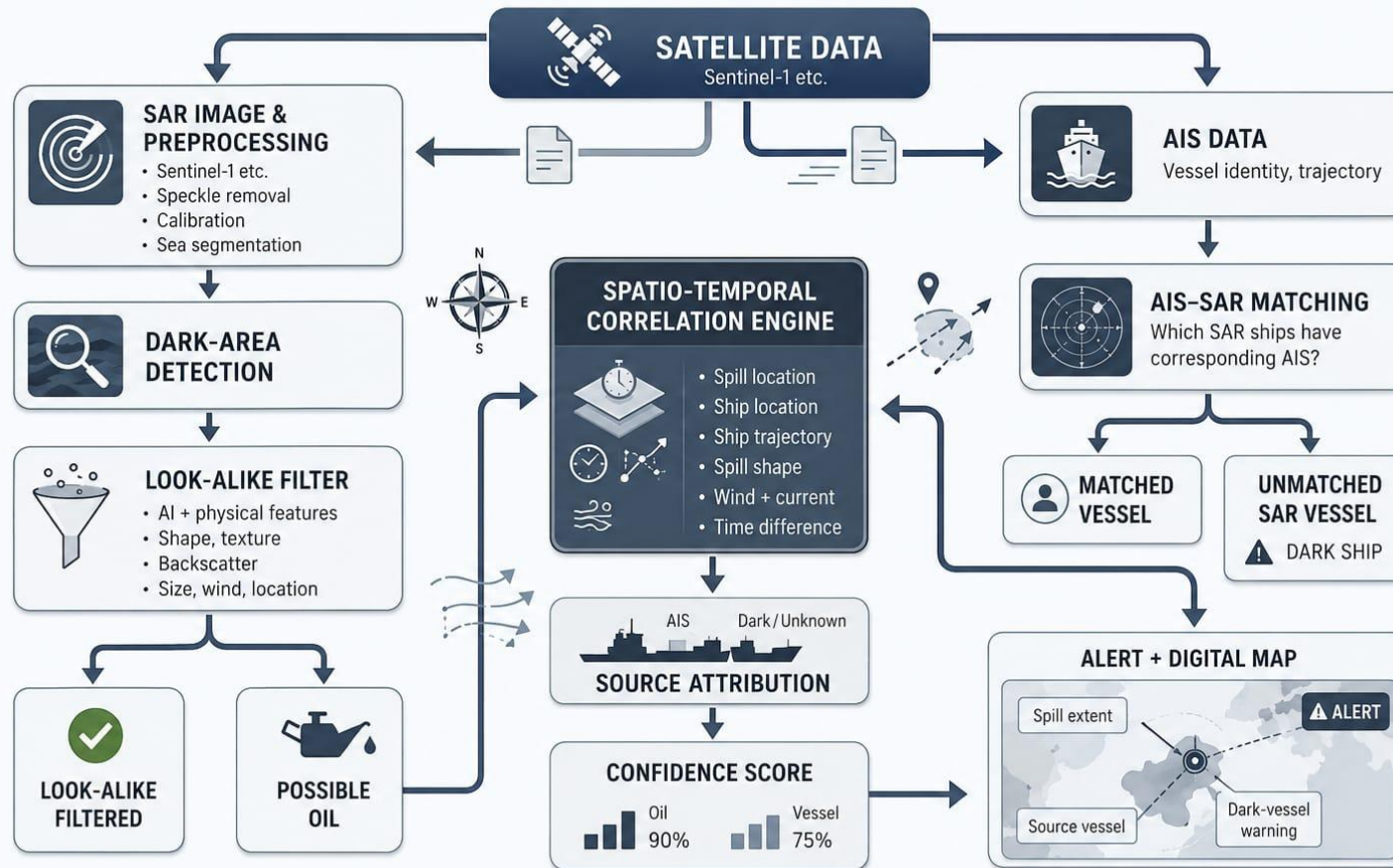




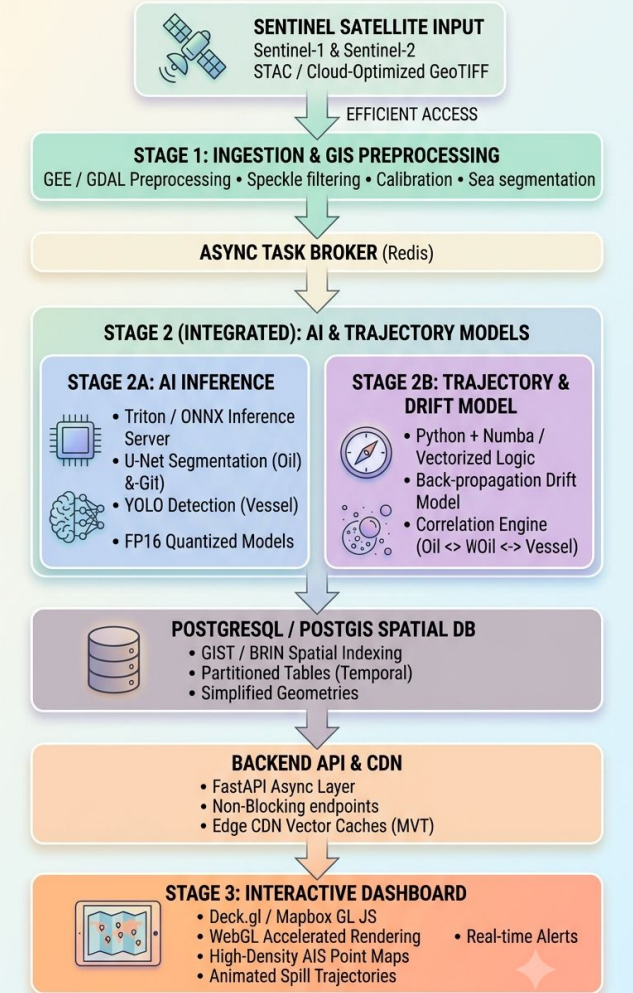
TECHNICAL APPROACH



INTEGRATED MARITIME SPILL DETECTION & SOURCE ATTRIBUTION FLOW

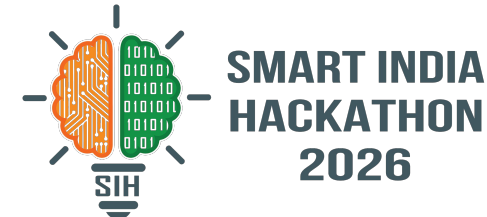


HIGH-EFFICIENCY MARITIME SPILL DETECTION ARCHITECTURE





FEASIBILITY AND VIABILITY



Why This Can Realistically Be Built

Technical Feasibility

- ❖ Sentinel-1 SAR data is free and openly available via Copernicus, with ~6–12 day revisit over Indian coastal waters.
- ❖ Open SAR oil-spill datasets exist for training/benchmarking segmentation models.
- ❖ Public AIS aggregators (AIS Hub, MarineTraffic API) supply vessel-track data for attribution.
- ❖ Reference systems (e.g. EMSA CleanSeaNet) already push pollution alerts within ~30 minutes of image acquisition — a proven benchmark to aim for.

Economic Feasibility

Parameter	Manual / Aerial Patrol	Proposed System
Coverage per pass	Narrow strip along patrol route	Full SAR swath, ~250 km wide
Operating conditions	Limited by weather / daylight	All-weather, day & night
Attribution to vessel	Requires witnesses / manual review	Automated via AIS cross-check
Running cost	High — fuel, crew, aircraft/vessel hours	Low — free satellite data + compute

Operational & Data Viability

- ❖ Small spills under ~7 tonnes are the hardest to catch manually, yet may account for up to 80% of global spill incidents — the automation gap is real.
- ❖ SAR works day/night and through cloud cover, unlike optical satellite imagery or manual patrols.
- ❖ Dashboard output (map + vessel ID + estimated area) is directly usable by Coast Guard / port control workflows.

Why It Matters for India

- ❖ India has a coastline of roughly 7,500 km and an EEZ of about 2.15 million km² to monitor.
- ❖ A large share of global seaborne oil trade transits Indian Ocean waters, raising spill risk near ports and shipping lanes.
- ❖ Manual/aerial patrols cover only a fraction of this area — satellite-based monitoring scales where patrols cannot.

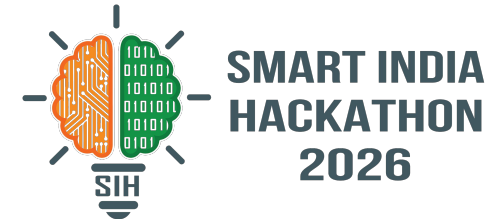


In the 2010s, there were 63 spills of 7 tonnes and over, resulting in 164,000 tonnes of oil lost.

In the 2020s to date, there have been 43 spills of 7 tonnes and over, resulting in 42,000 tonnes of oil lost.

Oil spills devastate the biosphere by coating habitats in toxic petroleum, which suffocates aquatic life and poisons the food chain

IMPACT AND BENEFITS



- Taking into account environmental features like wind speed/direction, ocean currents wave conditions temperature and rainfall can largely affect the capacity of SAR to predict oil slicks with accuracy.
- Studies show that wind speed/direction incorporation gives false alarm reduction ranging from 14% to 34% compared to mono-modal models.

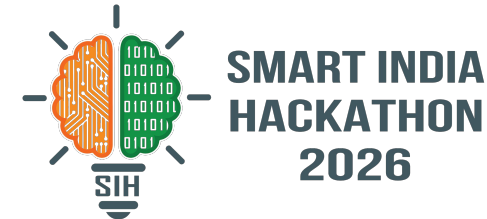
***Target:* 10–20% improvement in overall oil-spill identification reliability and a significant reduction in false positives through multi-source SAR + environmental-data fusion.**

- Sentinel-1 can detect ships even when they aren't broadcasting AIS.
- The European Space Agency specifically describes the capability of SAR and AIS integration for identifying so-called "**dark**" vessels.

***TARGET:* 15-35% improvement in probabilistic attribution of ships along with reduction in false attribution / missed-vessel cases through SAR–AIS cross-validation.**



RESEARCH AND REFERENCES



KEY REFERENCES

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- ✓ Zhao, J. et al., "Tracing illegal oil discharges from vessels using SAR and AIS in the Bohai Sea, China," Marine Pollution Bulletin, 2021.
- ✓ EMSA (European Maritime Safety Agency), "How does SAR detection work? Satellite-based Services," emsa.europa.eu.
- ✓ ICEYE, "Oil Spills SAR Data Use Case," iceye.com.
- ✓ Krestenitis, M. et al., "Full-Scale Aggregated MobileUNet for SAR Oil Spill Detection," 2024.
- ✓ Sentinel Online / ESA, "Sentinel-1 Supports Detection of Illegal Oil Spills," sentinel.esa.int.
- ✓ Case study: C-band Sentinel-1A SAR data for oil-spill detection off Chennai, East Coast of India, 2017 event, Marine Pollution Bulletin.

DATA SOURCES

